

Grouped Decision-Focused Model Averaging for System-Wide Capacity Commitment: Evidence from the U.S. Power Grid

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Abstract

System planners who coordinate many operating units must commit resources before demand is known, and they typically have several forecasting systems for each unit. We propose *grouped decision-focused model averaging* (GDMA), which combines the candidate forecasts of many units into commitment rules by jointly learning (i) a small number of latent segments of units and (ii) segment-specific combination weights, chosen to minimise the realised decision cost with unit-specific cost asymmetries rather than forecast error. A soft version shrinks each unit’s own weights towards those of its segment and nests pooled weights, unit-specific weights and shrinkage as special cases. We derive a finite-sample oracle inequality that separates the cost of learning segment memberships from the cost of learning weights, show that GDMA is asymptotically optimal and attains the risk of the infeasible benchmark with unit-specific optimal weights when segments exist, and prove that it recovers the segments with probability approaching one. Monte Carlo experiments confirm these properties. We apply the method to day-ahead capacity commitment for 46 U.S. balancing authorities, which together serve 97% of lower-48 electricity demand, using public hourly data from 2019 to September 2026. Over six and a half years of out-of-sample operation, soft GDMA lowers system-wide commitment cost by 18–20% relative to committing on the operators’ official forecasts, and by 3–16% relative to combining the same forecasts for accuracy first; it is the cheapest or statistically tied with the cheapest method for every estimation window. The learned segments separate operators whose own forecasts carry information beyond recent demand from those for whom machine-learned corrections suffice, a distinction invisible in the organisational structure of the grid.

Keywords: model averaging; forecast combination; decision-focused learning; latent groups; newsvendor; electricity grid; capacity commitment

1 Introduction

Large infrastructure systems are managed through many local decision units that commit resources before uncertainty resolves. The U.S. electric grid is an extreme example. Its lower 48 states are operated by several dozen balancing authorities—independent system operators, federal power marketing agencies, investor-owned and municipal utilities—each of which must, every day, commit enough generating capacity to meet the next day’s hourly demand. Committing too little forces expensive real-time purchases, the activation of reserves or, in the worst case, curtailed load; committing too much means paying for capacity that sits idle. The cost of

a shortfall is not the same everywhere: an operator embedded in a dense web of interconnections can import from its neighbours, while an isolated one cannot. Winter Storm Uri in February 2021, when the weakly interconnected Texas system shed load for days while better-connected neighbours coped, showed how much this asymmetry matters (Busby et al., 2021).

Every balancing authority publishes a day-ahead demand forecast, and a planner can build many more commitment rules from persistence, calendar profiles, statistical models and machine learning. None dominates everywhere, which makes combining them attractive. How to combine them is less obvious. The forecast-combination and model-averaging literature, starting with Bates and Granger (1969) and developed into an asymptotic theory by Hansen (2007), Hansen and Racine (2012), Zhang et al. (2013) and Zhang and Liu (2023), offers two standard options when there are many series. One can estimate combination weights for each series separately, which respects heterogeneity but produces noisy weights when histories are short—the root of the “forecast-combination puzzle” that simple averages often beat estimated weights (Smith and Wallis, 2009; Claeskens et al., 2016). Or one can pool all series and estimate one set of weights, which is precise but imposes the same policy on units that face different conditions. Both options also judge a combination by its statistical accuracy, even though the planner is judged by the cost of the resulting decisions. The operations-management literature has shown that learning directly for the decision—the “decision-focused” or “predict-then-optimize” perspective—can be much better than first forecasting accurately and then optimising (Ban and Rudin, 2019; Bertsimas and Kallus, 2020; Elmachtoub and Grigas, 2022), but this literature has not addressed the combination of existing forecasts across many heterogeneous units.

This paper proposes *grouped decision-focused model averaging* (GDMA), a new method designed for this setting. GDMA assumes that the decision units fall into a small number of latent segments whose members are best served by the same combination of candidate rules, and estimates the segments and the segment-specific weights jointly by minimising the realised system-wide decision cost. The number of segments is chosen by a time-ordered hold-out rule. Each unit keeps its own cost parameters and its own candidate safety margins, so segment members share a *rule* but not a commitment level. Pooled weights and unit-specific weights are the two extreme cases ($G = 1$ and $G = N$).

We make four contributions.

1. **Method.** We formulate the joint estimation of latent segments and combination weights under an asymmetric, unit-specific decision loss, and give a k -means-type algorithm with regret-based seeding in which each step is a small convex program on the simplex (Section 3). To the best of our knowledge neither the model-averaging literature, nor the decision-focused learning literature, nor the latent-group panel literature (Bonhomme and Manresa, 2015; Su et al., 2016) has proposed this estimator.
2. **Theory.** We prove a finite-sample oracle inequality that splits the estimation error into a term for learning memberships ($\log G/T$ per unit) and a term for learning weights ($GM \log(NT)/(NT)$), which formalises why grouping beats unit-specific weights when histories are short. We show that GDMA is asymptotically optimal and, under a segment structure, attains the risk of the infeasible benchmark in which every unit knows its own optimal weights; that it recovers the segments with probability approaching one; and that

on this event it coincides exactly with an estimator told the segments (Section 4).

3. **Evidence.** In Monte Carlo experiments GDMA is the most cost-effective feasible method when segments exist, loses little when they do not, and recovers the segments far more accurately than clustering noisy unit-level weights (Section 5).
4. **Application.** Using the complete public record of hourly demand and official day-ahead forecasts of 46 U.S. balancing authorities from 2019 to September 2026 (EIA Form 930), we evaluate day-ahead capacity commitment over more than six years of out-of-sample operation, including Winter Storms Uri and Elliott and the summer heat waves of 2023–2025 (Section 6). The results quantify how much system-wide commitment cost a planner could save by combining forecasts in a decision-focused, segment-aware way, and which operators should share a policy.

The paper speaks directly to the theme of this special issue: it develops a statistical method motivated by a concrete management problem, establishes its theoretical guarantees, and demonstrates its value for a decision of national scale using new, openly available data. Code and data-construction scripts are available from the author.

2 Related literature and the research gap

Model averaging and forecast combination. Since [Bates and Granger \(1969\)](#), combining forecasts has been one of the most reliable ways to improve predictive accuracy; [Timmermann \(2006\)](#) and [Wang et al. \(2023\)](#) review five decades of work. Frequentist model averaging chooses weights by minimising an estimate of risk—a Mallows criterion ([Hansen, 2007](#)), leave-one-out or K -fold cross-validation ([Hansen and Racine, 2012](#); [Zhang and Liu, 2023](#)), or forward validation for time series ([Zhang and Zhang, 2023](#))—and establishes asymptotic optimality, i.e. that the chosen weights attain the lowest attainable risk. The framework now covers dependent data ([Zhang et al., 2013](#)), quantile losses ([Lu and Su, 2015](#)), general losses ([Yu et al., 2024](#)), time-varying weights ([Sun et al., 2021](#)), covariate-dependent weights ([Gong et al., 2026](#)), covariate shift ([Zhang et al., 2026](#)) and prediction intervals ([Qu et al., 2025](#)). The forecast-combination puzzle—that equal weights frequently beat estimated weights—is attributed mainly to estimation error in the weights ([Smith and Wallis, 2009](#); [Claeskens et al., 2016](#)); remedies shrink estimated weights towards equal weights ([Diebold and Shin, 2019](#)) or restrict them ([Zou et al., 2025](#)). When many series must be forecast, weights are either estimated series by series or learned from series features (FFORMA, [Montero-Manso et al., 2020](#)); [Aiolfi and Timmermann \(2006\)](#) cluster *forecasting models* by past performance. All of these approaches either treat each target series separately or impose one rule on all of them, and none groups the *target units*.

Decision-focused learning. Operations research has long recognised that the loss that matters is the decision cost ([Granger, 1969](#)). The data-driven newsvendor literature learns order quantities directly from data ([Levi et al., 2015](#); [Ban and Rudin, 2019](#)); prescriptive analytics maps covariates to decisions ([Bertsimas and Kallus, 2020](#)); and “smart predict-then-optimise” trains predictors on decision regret ([Elmachtoub and Grigas, 2022](#)); see [Mandi et al. \(2024\)](#)

Table 1: Positioning of GDMA relative to existing approaches.

Approach	Combines candidates	Decision- focused loss	Shares strength across units	Heterogeneous weights
Per-series combination / JMA / forward validation	✓			✓
Pooled combination	✓		✓	
Quantile / general-loss model averaging	✓	✓		✓
Shrinkage towards equal or pooled weights	✓		partial	✓
Feature-based weights (FFORMA)	✓		✓	✓
Data-driven newsvendor / predict-then-optimize		✓	✓	
Latent-group panel models			✓	✓
GDMA (this paper)	✓	✓	✓	✓

for a survey. In energy forecasting, Wang et al. (2019) combine probabilistic load forecasts by minimising pinball loss on the simplex, one series at a time. This literature trains single models or combines forecasts for one unit; it does not exploit cross-sectional structure when many heterogeneous units face related decisions.

Latent group structures. Panel econometrics models unobserved heterogeneity through latent groups estimated by k -means-type algorithms (Lin and Ng, 2012; Bonhomme and Manresa, 2015; Ando and Bai, 2016) or classifier-Lasso (Su et al., 2016). These methods group *regression parameters* under squared loss and focus on the parameters themselves, not on combining the outputs of existing forecasting systems under an asymmetric decision loss.

The gap. Table 1 summarises the position of GDMA. No existing method simultaneously (a) combines a library of existing candidate rules, (b) chooses weights by the downstream decision cost with unit-specific cost asymmetries, and (c) shares statistical strength across many decision units through latent segments whose number is chosen from data, with guarantees on optimality and segment recovery. Filling this gap matters most when windows are short—because systems change and old data are stale—and units are many, which is exactly the situation of system-wide infrastructure planning.

3 Grouped decision-focused model averaging

3.1 The system-wide commitment problem

A planner is responsible for N decision units (in our application the balancing authorities of the U.S. grid). In period t unit i must commit a quantity q_{it} before demand y_{it} is realised. Committing too little costs $u_i > 0$ per unit of shortfall (emergency purchases, reserve activation or, in the limit, curtailed load); committing too much costs $o_i > 0$ per unit of excess (idle capacity that was paid for). The period cost is the newsvendor cost

$$C_i(q, y) = u_i (y - q)^+ + o_i (q - y)^+ = (u_i + o_i) \rho_{\tau_i}(y - q), \quad \tau_i = \frac{u_i}{u_i + o_i}, \quad (1)$$

where $\rho_\tau(r) = r(\tau - \mathbb{1}\{r < 0\})$ is the check function. The system cost that the planner cares about is $\sum_i \sum_t C_i(q_{it}, y_{it})$; under (1) the cost-minimising commitment is the conditional τ_i -quantile of demand (Gneiting, 2011). Critical ratios τ_i differ across units because the consequences of a shortfall differ: an operator that is tightly interconnected with its neighbours can import at moderate cost, while an isolated one cannot.

For every unit a library of M candidate commitment rules is available. Candidate m proposes $q_{it}^{(m)}$, computed from information available before period t ; we write $\mathbf{q}_{it} = (q_{it}^{(1)}, \dots, q_{it}^{(M)})^\top$. The candidates may come from any source: an operator’s own published forecast, simple persistence rules, statistical models or machine-learning models, each turned into a commitment by adding its own τ_i -quantile residual margin. The planner combines them linearly,

$$q_{it}(\mathbf{w}) = \mathbf{q}_{it}^\top \mathbf{w}, \quad \mathbf{w} \in \Delta_M = \left\{ \mathbf{w} \in \mathbb{R}^M : w_m \geq 0, \sum_m w_m = 1 \right\}.$$

The simplex constraint keeps the combined commitment inside the range spanned by the candidates, which is what makes the combination interpretable and auditable by operators, and it is known to stabilise estimated weights (Zou et al., 2025).

3.2 Three ways to estimate the weights

Let the estimation window contain periods $t = 1, \dots, T$ for which both candidate commitments and realised demand are observed (a forward-validation sample in the terminology of Zhang and Zhang, 2023). Write $\hat{R}_i(\mathbf{w}) = T^{-1} \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}, y_{it})$ for the empirical cost of unit i .

Unit-specific weights. $\hat{\mathbf{w}}_i = \arg \min_{\mathbf{w} \in \Delta_M} \hat{R}_i(\mathbf{w})$. This respects heterogeneity but estimates $M - 1$ free parameters from T serially dependent observations per unit; it is the decision-focused counterpart of per-series forecast combination, and suffers from the same estimation noise that underlies the forecast-combination puzzle (Smith and Wallis, 2009; Claeskens et al., 2016).

Pooled weights. $\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \sum_i \hat{R}_i(\mathbf{w})$. This uses all NT observations but forces a single policy on units that may need different ones.

Grouped weights (proposed). We posit that units fall into a small number G of latent segments whose members are well served by the same combination. The GDMA estimator solves

$$(\hat{\gamma}, \hat{\mathbf{W}}) = \arg \min_{\gamma \in \{1, \dots, G\}^N, \mathbf{W} = (\mathbf{w}_1, \dots, \mathbf{w}_G) \in \Delta_M^G} \sum_{i=1}^N \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}_{\gamma_i}, y_{it}), \quad (2)$$

and unit i commits $\mathbf{q}_{i,T+1}^\top \hat{\mathbf{w}}_{\hat{\gamma}_i}$. $G = 1$ gives pooled weights and $G = N$ gives unit-specific weights, so the number of segments indexes the whole range between the two extremes. Three features separate (2) from existing proposals. First, the criterion is the decision cost in the planner’s own units (money or energy), so units are grouped by *how they should act*, not by how their forecasts look. Second, memberships and weights are estimated jointly; a two-step procedure that clusters noisy unit-specific weights and then re-estimates is a natural but inferior alternative (Section 5). Third, because each unit keeps its own critical ratio and its own candidate margins, units in one segment share a combination rule but not a commitment level.

Algorithm 1 GDMA for a fixed number of segments G

Require: candidate commitments $\{q_{it}\}$, demand $\{y_{it}\}$, costs $\{u_i, o_i\}$, G , number of starts S

- 1: compute unit-specific weights $\hat{w}_i$ and the regret matrix $D_{ij} = \hat{R}_i(\hat{w}_j) - \hat{R}_i(\hat{w}_i)$
 - 2: **for** $s = 1, \dots, S$ **do**
 - 3: draw seeds $j_1, \dots, j_G$ by k -means++ on D and set $w_g \leftarrow \hat{w}_{j_g}$
 - 4: **repeat**
 - 5: $\gamma_i \leftarrow \arg \min_g \hat{R}_i(w_g)$ for all i $\triangleright$ assignment step
 - 6: re-seed any empty segment with the unit of largest regret
 - 7: $w_g \leftarrow \arg \min_{w \in \Delta_M} \sum_{i: \gamma_i = g} \hat{R}_i(w)$ for all g $\triangleright$ pooled convex step
 - 8: **until** memberships do not change
 - 9: **end for**
 - 10: **return** the solution with the lowest objective (2)
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3.3 Computation

Problem (2) is a mixed-integer problem, but it has the structure of k -means with a non-Euclidean, data-dependent distortion, as in the grouped fixed-effects estimator of Bonhomme and Manresa (2015). Given memberships, the weights of each segment solve a pooled convex problem on the simplex; given weights, each unit joins the segment whose weights give it the lowest in-window cost. Algorithm 1 alternates the two steps. The weight step is a linear program; for speed we minimise a softplus-smoothed check function, $\rho_r^\varepsilon(r) = \tau r + \varepsilon \log(1 + e^{-r/\varepsilon})$, with bandwidth ε equal to 2% of the unit’s typical decision error, by sequential quadratic programming. On our data the smoothed solution is within 0.01% of the exact linear-programming optimum in cost. Each iteration weakly decreases the objective, so the algorithm converges to a local solution in finitely many steps; we run several starts and keep the best. Starting values are drawn by a k -means++ rule (Arthur and Vassilvitskii, 2007) in *regret space*: the next seed unit is drawn with probability proportional to the extra cost that units would incur if forced to use the closest existing seed’s own weights. In rolling applications the memberships of the previous window provide an additional warm start.

3.4 Soft segments

Segments need not be exact. When units within a segment are similar but not identical, a unit may benefit from a rule that sits between its own estimated weights and those of its segment. *Soft GDMA* uses

$$\hat{w}_i^{\text{soft}} = (1 - \kappa) \hat{w}_i + \kappa \hat{w}_{\gamma_i}, \quad \kappa \in [0, 1], \quad (3)$$

where $\hat{w}_i$ are the unit-specific weights and $\hat{w}_{\gamma_i}$ the GDMA weights of the unit’s segment. Because the simplex is convex, $\hat{w}_i^{\text{soft}} \in \Delta_M$. The family nests every benchmark: $\kappa = 1$ is GDMA, $\kappa = 0$ is unit-specific weights, $G = 1$ is shrinkage towards pooled weights (a decision-focused analogue of Diebold and Shin, 2019), and $G = 1, \kappa = 1$ is pooled weights. The number of segments is chosen exactly as for GDMA (next subsection); given G , κ is chosen on the same hold-out block from $\{0, 0.25, 0.5, 0.75, 1\}$. Choosing G first keeps the search small, which matters because hold-out costs are noisy when windows are short. Segment centres play the role of data-driven prior means, in the spirit of empirical Bayes shrinkage towards group-specific targets.

3.5 Choosing the number of segments

We split the estimation window into an early block (the first two thirds) and a late block. For each $G \in \{1, \dots, G_{\max}\}$ we estimate (2) on the early block and record every unit's cost on the late block under its estimated membership and weights. Let $G^\dagger$ minimise the total late-block cost. Following the one-standard-error rule (Hastie et al., 2009), we choose the smallest G whose total late-block cost exceeds that of $G^\dagger$ by no more than one standard error of the summed unit-level cost differences, and then re-estimate on the full window. The rule respects time order and prefers parsimonious segmentations, which are easier to explain to the operators who would have to adopt them.

4 Theoretical properties

This section studies the global minimiser of (2) for a given G . We show that (i) its expected decision cost is close to that of the best grouped policy, with an estimation error that separates the price of learning memberships from the price of learning weights; (ii) it is asymptotically optimal and, under a latent segment structure, attains the risk of the infeasible unit-specific oracle; and (iii) it recovers the segments with probability approaching one, in which case its weights coincide with those of an estimator that is told the segments. Proofs are in Appendix A.

4.1 Setting and assumptions

Let $\mathbf{z}_{it} = (y_{it}, \mathbf{q}_{it})$. Define the population risk of unit i , $R_i(\mathbf{w}) = \mathbb{E} C_i(\mathbf{q}_{it}^\top \mathbf{w}, y_{it})$, and for a grouped policy $(\gamma, \mathbf{W})$ the average risk and its empirical analogue

$$R(\gamma, \mathbf{W}) = \frac{1}{N} \sum_{i=1}^N R_i(\mathbf{w}_{\gamma_i}), \quad \hat{R}(\gamma, \mathbf{W}) = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}_{\gamma_i}, y_{it}).$$

Let $\mathcal{P}_G = \{1, \dots, G\}^N \times \Delta_M^G$ and $R_G^* = \inf_{\mathcal{P}_G} R$.

Assumption 1 (Stationarity and boundedness). *For each i , $\{\mathbf{z}_{it}\}_t$ is strictly stationary, $|y_{it}| \leq B$ and $\|\mathbf{q}_{it}\|_\infty \leq B$ almost surely, and $\max_i(u_i + o_i) \leq \bar{a}$.*

Stationarity refers to the candidate commitments as well as demand; it holds, for instance, when every candidate is computed from a rolling window of fixed length of a stationary process. Boundedness is natural once demand is normalised by a unit's scale. Under Assumption 1 each loss lies in $[0, L]$ with $L = 2\bar{a}B$ and is Lipschitz in the weights: $|C_i(\mathbf{q}^\top \mathbf{w}, y) - C_i(\mathbf{q}^\top \mathbf{w}', y)| \leq \bar{a}B \|\mathbf{w} - \mathbf{w}'\|_1$.

Assumption 2 (Concentration). *There are constants $c > 0$ and an effective sample length $T_e \leq T$ such that, for every fixed $(\gamma, \mathbf{W}) \in \mathcal{P}_G$ and every $x > 0$,*

$$\Pr(|\hat{R}(\gamma, \mathbf{W}) - R(\gamma, \mathbf{W})| > x) \leq 2 \exp(-c N T_e x^2 / L^2),$$

and, for every i and fixed $\mathbf{w}$, $\Pr(|\hat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| > x) \leq 2 \exp(-c T_e x^2 / L^2)$.

Assumption 2 is a high-level condition. It holds with $T_e = T$ when observations are independent (Hoeffding’s inequality) and, when units are independent and each $\{\mathbf{z}_{it}\}_t$ is geometrically β -mixing, with $T_e = T/(C \log T)$ by the blocking argument of Yu (1994) or the Bernstein inequality of Merlevède et al. (2011). Cross-sectional dependence through common shocks (weather, in our application) reduces the effective number of independent units; the results continue to hold with N replaced by an effective cross-sectional size.

4.2 An oracle inequality

Theorem 1 (Oracle inequality). *Let Assumptions 1–2 hold and let $(\hat{\gamma}, \hat{\mathbf{W}})$ be a global minimiser of (2). For any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$R(\hat{\gamma}, \hat{\mathbf{W}}) \leq R_G^* + 2L \sqrt{\frac{GM \log(3NT_e) + N \log G + \log(2/\delta)}{c NT_e}} + \frac{4aB}{NT_e}. \quad (4)$$

The square-root term has a transparent decomposition. Per unit, learning the segment membership costs $\log G/T_e$, while learning the segment weights costs $GM \log(NT_e)/(NT_e)$. The same argument applied to unit-specific weights ($G = N$) gives $M \log(NT_e)/T_e$, and applied to pooled weights ($G = 1$) gives $M \log(NT_e)/(NT_e)$ but a larger approximation error $R_1^* - R_N^*$. Grouping therefore replaces the M per-unit parameters of unit-specific weights by a single discrete label that costs $\log G$ rather than $M \log(NT_e)$ to learn. This is the formal sense in which GDMA resolves the tension between the noise of unit-specific weights and the bias of pooled weights: with $M = 7$ candidates, $N = 46$ units, $G = 4$ segments and an effective length of $T_e = 28$ (one observation per day of a four-week window), the variance proxy under the square root is 0.23 for GDMA against 2.07 for unit-specific weights, so the bound is about three times tighter, while the approximation error of GDMA is zero whenever the segment structure is correct.

4.3 Asymptotic optimality

Following the model-averaging literature (Hansen, 2007; Hansen and Racine, 2012; Zhang and Liu, 2023), call a weight choice asymptotically optimal if the ratio of its risk to the lowest attainable risk in its class tends to one.

Corollary 1 (Asymptotic optimality). *Let the assumptions of Theorem 1 hold, with G and M possibly growing, and suppose $R_G^* \geq \kappa > 0$. If $GM \log(NT_e)/(NT_e) \rightarrow 0$ and $\log G/T_e \rightarrow 0$, then $R(\hat{\gamma}, \hat{\mathbf{W}})/R_G^* \rightarrow_p 1$. If, in addition, there are memberships γ^0 with $G_0 \leq G$ segments such that $\arg \min_{\mathbf{w}} R_i(\mathbf{w})$ depends on i only through γ_i^0 , then $R_G^* = R_N^* = N^{-1} \sum_i \min_{\mathbf{w}} R_i(\mathbf{w})$ and*

$$\frac{R(\hat{\gamma}, \hat{\mathbf{W}})}{N^{-1} \sum_i \min_{\mathbf{w} \in \Delta_M} R_i(\mathbf{w})} \rightarrow_p 1.$$

The second statement is the relevant one for practice: under a segment structure, GDMA attains the risk of the infeasible benchmark in which every unit knows its own optimal weights, although it estimates only $G(M - 1)$ continuous parameters. Unit-specific weights attain the same limit only if $M \log(NT_e)/T_e \rightarrow 0$, a much stronger requirement when windows are short

because the system is non-stationary and old data are of limited use. The condition $R_G^* \geq \kappa$ excludes the degenerate case of perfect foresight, which never occurs in commitment problems.

4.4 Recovery of the segments

Assumption 3 (Segment structure). (a) There are $\gamma^0 \in \{1, \dots, G\}^N$ and distinct $\mathbf{w}_1^0, \dots, \mathbf{w}_G^0 \in \Delta_M$ such that, for all i and all $\mathbf{w} \in \Delta_M$, $R_i(\mathbf{w}) - R_i(\mathbf{w}_{\gamma_i^0}^0) \geq \lambda \|\mathbf{w} - \mathbf{w}_{\gamma_i^0}^0\|_2^2$ for some $\lambda > 0$. (b) $\min_{g \neq h} \|\mathbf{w}_g^0 - \mathbf{w}_h^0\|_2 \geq d > 0$ and every segment contains at least πN units, $\pi > 0$.

Part (a) is a quadratic-growth condition. It holds when the conditional density of demand given the candidates is bounded away from zero near the optimal commitment and the candidates are not collinear, because the Hessian of R_i is then $(u_i + o_i) \mathbb{E}[f_{y|q}(\mathbf{q}^\top \mathbf{w}) \mathbf{q} \mathbf{q}^\top]$, which is positive definite on the tangent space of the simplex. Part (b) requires segments to be genuinely different and non-negligible, as in [Bonhomme and Manresa \(2015\)](#).

Theorem 2 (Segment recovery and oracle equivalence). Let Assumptions 1–3 hold with G equal to the true number of segments, and let $N, T_e \rightarrow \infty$ with $\log N/T_e \rightarrow 0$. Then, up to a relabelling of segments, (i) $\max_g \|\hat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \rightarrow_p 0$; (ii) $\Pr(\hat{\gamma} \neq \gamma^0) \leq N C_M \exp(-c' T_e) + o(1) \rightarrow 0$ for constants $C_M, c' > 0$ that do not depend on N or T ; (iii) with probability approaching one, $\hat{\mathbf{W}}$ equals the infeasible estimator $\tilde{\mathbf{W}} = \arg \min_{\mathbf{W}} \hat{R}(\gamma^0, \mathbf{W})$ that knows the segments.

Part (iii) is an exact equivalence, not an asymptotic approximation: on the event that memberships are recovered, the joint minimiser and the known-segment minimiser are the same object. Any distributional result available for the known-segment estimator therefore transfers to GDMA without modification. The recovery condition $\log N/T_e \rightarrow 0$ is mild: the number of units may grow polynomially in the window length.

Soft GDMA. For fixed (G, κ) , R_i is convex in $\mathbf{w}$, so the risk of (3) satisfies $R_i(\hat{\mathbf{w}}_i^{\text{soft}}) \leq (1 - \kappa) R_i(\hat{\mathbf{w}}_i) + \kappa R_i(\hat{\mathbf{w}}_{\hat{\gamma}_i})$. Averaging over units, the excess risk of soft GDMA is at most the κ -weighted average of the excess risks of unit-specific weights and of GDMA, each controlled by Theorem 1. Under a segment structure, therefore, any κ bounded away from zero inherits the faster rate of GDMA for the κ share of the combination, while $\kappa < 1$ buys robustness when segments are only approximate. Choosing (G, κ) from a finite grid by hold-out adds only a $\log(|\mathcal{G}|)$ term to the hold-out error, where $|\mathcal{G}|$ is the grid size.

5 Monte Carlo evidence

PENDING

6 Day-ahead capacity commitment on the U.S. grid

6.1 Data

We use the complete public record of the U.S. Energy Information Administration’s Hourly Electric Grid Monitor (Form EIA-930; [U.S. Energy Information Administration, 2026](#)), downloaded on 29 September 2026. For every balancing authority (BA) in the lower 48 states the file

reports hourly demand, the BA’s own day-ahead demand forecast and total interchange with neighbouring BAs. We use the period 1 January 2019 to 26 September 2026 in UTC hours and exclude the fourteen regional aggregates. Following [Ruggles et al. \(2020\)](#), non-positive values and values outside $[0.3, 3]$ times a centred seven-day rolling median are treated as missing, and gaps of up to three hours are interpolated. We keep BAs with at least 97% valid demand, at least 93% valid official forecasts, mean demand above 50 MW and a median absolute official forecast error below 15% of median demand. Forty-six BAs remain (Table 6 in Appendix C); they range from PJM (mean demand 92.5 GW) to the City of Homestead (71 MW), include all seven independent system operators, and account for 97% of lower-48 demand. Ten BAs are dropped, mostly small systems with long gaps or implausible official forecasts; 0.27% of the retained BA-hours are missing.

6.2 Decision problem and cost calibration

Before each operating day d , BA i commits hourly capacity $q_{i,d,h}$, $h = 0, \dots, 23$. The information set contains realised demand up to the end of day $d-1$ and the BA’s published day-ahead forecast for day d . Excess commitment costs $o_i = 1$ per MWh, which normalises all costs to units of the per-MWh cost of idle committed capacity. Shortfalls cost $u_i = \tau_i / (1 - \tau_i)$, where the critical ratio τ_i reflects the BA’s ability to import. We measure interchange flexibility in the pre-sample year 2019 as the range between the 5th and 95th percentiles of hourly total interchange, divided by mean demand, and let τ_i decline linearly in the flexibility rank from 0.95 for the least flexible BA to 0.80 for the most flexible one. ERCOT, which is almost islanded, receives $\tau = 0.95$ ($u_i = 19$); the Chelan County PUD in the densely interconnected Pacific Northwest receives $\tau = 0.80$ ($u_i = 4$). This calibration is deliberately simple and transparent; Section 6.7 shows that the ranking of methods is the same under common critical ratios between 0.5 and 0.95.

6.3 Candidate rules

Seven candidate forecasts are produced for every BA, day and hour, each using only information available before the operating day: (1) the BA’s official day-ahead forecast; (2) persistence (same hour of $d-1$); (3) weekly persistence (same hour of $d-7$); (4) the mean of the same hour over $d-1, \dots, d-7$; (5) the mean of the same hour on $d-7$, $d-14$ and $d-21$; (6) a BA-and-hour-specific linear regression on the $d-1$ and $d-7$ values, the official forecast and weekday dummies, estimated on a rolling 56-day window and refitted weekly; and (7) a single global LightGBM model ([Ke et al., 2017](#)) across all BAs and hours, trained on lags, calendar variables and the official forecast, all scaled by the BA’s recent mean demand, refitted every 28 days on the preceding 365 days. Model (7) is the kind of machine-learning forecaster a central planner could run for every BA at low cost. Each forecast $f^{(m)}$ becomes a candidate commitment by adding its own safety margin,

$$q_{i,d,h}^{(m)} = f_{i,d,h}^{(m)} \{1 + \hat{Q}_{\tau_i}(e_i^{(m)})\},$$

where $\hat{Q}_{\tau_i}(e_i^{(m)})$ is the empirical τ_i -quantile of the relative errors of forecast m for BA i over the previous 28 days. The candidate rules are available from February 2020, so the evaluation

Table 2: Out-of-sample system-wide commitment cost, 1 April 2020 to 26 September 2026, relative to committing on the official day-ahead forecasts. W is the length of the estimation window. Bold: lowest cost in the column. Stars: Diebold–Mariano test of equal daily system cost against soft GDMA with Newey–West variance (7 lags); *, **, ***: $p < 0.10, 0.05, 0.01$.

Method	System cost relative to the official-forecast policy				Share of BAs improved ^a
	$W = 7$ days	$W = 14$ days	$W = 28$ days	$W = 56$ days	
Official forecast	1.0000***	1.0000***	1.0000***	1.0000***	0.00
Equal weights	1.5322***	1.5322***	1.5322***	1.5322***	0.13
Best single (per BA)	0.8490***	0.8289***	0.8245***	0.8234***	0.76
Pooled DF weights	0.8156	0.8110***	0.8076***	0.8040***	0.78
Per-BA DF weights	0.8497***	0.8221***	0.8069***	0.8008***	0.78
Shrinkage to pooled	0.8154	0.8054	0.8010	0.7963	0.80
Two-step k -means	0.8243	0.8077*	0.8041***	0.7981	0.83
Forecast-then-commit, per BA	0.9713***	0.8850***	0.8382***	0.8233***	0.65
Forecast-then-commit, grouped	0.9068***	0.8523***	0.8260***	0.8277***	0.67
GDMA	0.8163	0.8079***	0.8042***	0.7999***	0.78
GDMA, minimum hold-out rule	0.8227***	0.8112***	0.8034***	0.7978*	0.80
Soft GDMA	0.8166	0.8050	0.8000	0.7966	0.80

^a Share of the 46 BAs whose own cost is lower than under the official-forecast policy ($W = 28$).

period runs from 1 April 2020 to 26 September 2026: 2,342 days, 335 weekly re-estimations and 2.58 billion MWh of demand.

6.4 Competing methods

Every seven days each method re-estimates its combination from the most recent W days, $W \in \{7, 14, 28, 56\}$ (each day contributes 24 hourly observations per BA) and commits for the following seven days. Besides GDMA and soft GDMA ($G_{\max} = 6$) we consider: the official-forecast policy (candidate 1 alone), which is the status quo; equal weights over the seven candidates; the best single candidate per BA over the window; pooled decision-focused weights; per-BA decision-focused weights (the decision-focused analogue of per-series forward validation); shrinkage of per-BA weights towards pooled weights with the shrinkage intensity chosen on the hold-out (a decision-focused version of [Diebold and Shin, 2019](#)); two-step clustering, which runs k -means on the per-BA weight vectors and re-estimates pooled weights within clusters, with the number of clusters chosen by the same hold-out; and two *forecast-then-commit* methods, which estimate per-BA or grouped combination weights by minimising scaled squared forecast error and then add the τ_i -quantile of the combined forecast’s relative errors. The last two isolate the value of estimating the combination for the decision rather than for accuracy.

6.5 Results

Table 2 contains the main result. Four findings stand out.

First, deciding for the decision pays. Soft GDMA lowers the system-wide commitment cost by 18–20% relative to the status quo of committing on the official forecasts, and every decision-focused combination method lowers it by at least 15%; all reductions are significant at the 1% level. The forecast-then-commit methods, which combine the same seven forecasts for accuracy

and then add a quantile margin, save only 3–18%, and they are significantly more expensive than soft GDMA for every window length. The gap is largest when windows are short ($W = 7$: 0.971 and 0.907 against 0.817), because accuracy-optimal weights are chosen without regard to the asymmetric costs and the margin estimated from one week of combined errors is noisy. Equal weights are the worst choice: they give weight to the naive calendar rules and raise cost by 53%.

Second, neither extreme is right. Per-BA weights are expensive when windows are short (0.850 at $W = 7$, 4.2% above pooled weights), pooled weights are expensive when windows are long (0.804 at $W = 56$, 0.9% above soft GDMA), and the per-window ranking of the two reverses between $W = 14$ and $W = 28$. This is the bias–variance trade-off of Theorem 1 in action.

Third, soft GDMA is the most reliable method. It has the lowest system cost for $W = 14$ and $W = 28$ and is within 0.1% of the best method for $W = 7$ and $W = 56$; no method is significantly cheaper than soft GDMA for any window, while it is significantly cheaper than pooled weights, per-BA weights, the forecast-then-commit methods and plain GDMA for most windows. Its closest competitor is decision-focused shrinkage towards pooled weights, which is the special case $G = 1$ of soft GDMA and is statistically indistinguishable from it; the two coincide exactly in the 57–68% of weeks (depending on W) in which the hold-out rule selects a single segment. Plain GDMA improves significantly on pooled weights for $W \geq 14$ and on per-BA weights for $W \leq 14$, but forcing all members of a segment onto identical weights costs 0.4–0.5% relative to soft segments when $W \geq 14$. Choosing G by the plain hold-out minimum instead of the one-standard-error rule is worse for short windows (0.823 against 0.816 at $W = 7$), confirming that parsimony matters when hold-out costs are noisy.

Fourth, the savings are economically large. At $W = 28$ the official-forecast policy costs 0.0761 units of idle-capacity cost per MWh of demand and soft GDMA 0.0610, a saving of 0.0151 units per MWh. Over the 4.0 billion MWh of annual demand in our panel, and valuing idle committed capacity at an illustrative \$10 per MWh, this is roughly \$600 million a year for the system as a whole, obtained by re-weighting forecasts that the operators already produce. The saving is not driven by a few operators: 80% of BAs are individually better off than under their own official forecast (last column of Table 2).

Table 3 and Figure 1 confirm the picture. With every BA weighted equally rather than by its size, soft GDMA and shrinkage are again the two cheapest methods, so the results are not an artefact of the three largest ISOs. The decision-focused methods hit the target shortfall frequency closely (11.5–12.0% against a target of 12.5%, the average of $1 - \tau_i$), whereas the official-forecast policy and the forecast-then-commit methods fall short too often (13.5–14.5%). The cumulative saving builds up steadily over six years rather than in a few episodes, and the ordering of the methods is stable after the first year.

6.6 Segments

GDMA uses one segment in about two thirds of the weeks and two or more in the rest; multi-segment weeks occur in every year from 2021 onwards (Figure 2, left). The co-membership matrix (right) reveals two stable segments. The first contains 17 BAs, among them PJM, CAISO, ISO New England, Florida Power & Light, Arizona Public Service, Nevada Power and both Pacifi-

Table 3: Secondary metrics for $W = 28$ days.

Method	Scale-free cost ^b (rel. to EW)	Shortfall frequency	DM statistic vs soft GDMA ^c
Official forecast	0.7374	0.135	27.51
Equal weights	1.0000	0.083	35.86
Best single (per BA)	0.6274	0.138	7.62
Pooled DF weights	0.6212	0.117	5.32
Per-BA DF weights	0.6166	0.116	3.78
Shrinkage to pooled	0.6137	0.115	1.22
Two-step k -means	0.6169	0.120	2.87
Forecast-then-commit, per BA	0.6311	0.145	6.91
Forecast-then-commit, grouped	0.6294	0.137	7.26
GDMA	0.6194	0.119	4.49
GDMA, minimum hold-out rule	0.6175	0.120	4.64
Soft GDMA	0.6140	0.116	—
Target (mean τ_i across BAs)		0.125	

^b Each BA’s cost divided by its mean hourly demand, averaged over BAs with equal weight, relative to equal weights. ^c Diebold–Mariano statistic; positive values: the method has higher daily system cost than soft GDMA.

Table 4: System cost during extreme events relative to the official-forecast policy ($W = 28$): Winter Storm Uri (10–20 February 2021), Winter Storm Elliott (22–27 December 2022) and the heat wave of 15 July–15 August 2023.

Method	Uri	Elliott	Heat wave 2023
Official forecast	1.000	1.000	1.000
Equal weights	1.133	2.534	1.366
Best single (per BA)	0.881	1.264	0.796
Pooled DF weights	0.842	1.318	0.778
Per-BA DF weights	0.848	1.184	0.790
Shrinkage to pooled	0.845	1.210	0.779
Two-step k -means	0.842	1.239	0.780
Forecast-then-commit, per BA	0.973	1.584	0.796
Forecast-then-commit, grouped	0.946	1.481	0.785
GDMA	0.842	1.318	0.782
GDMA, minimum hold-out rule	0.841	1.221	0.788
Soft GDMA	0.845	1.210	0.780

Corp areas; their combinations rely almost entirely on the learned models (weights of 0.38 on the hourly regression and 0.47 on LightGBM) and put only 0.12 directly on the official forecast. The second contains 28 BAs, including MISO, ERCOT, the Southern Company, NYISO, TVA, Bonneville and Duke Energy, and puts 0.39 directly on the official forecast. Southwest Power Pool often forms a segment of its own. The two main segments have the same average critical ratio (0.874 and 0.873), so the segmentation is not a relabelling of cost asymmetries; nor does it follow geography or market design, since both contain ISOs and vertically integrated utilities from the East and the West. What separates them is how much the operator’s own forecast adds to what can be learned from recent demand—information that a central planner cannot read off the organisational chart of the grid, but that GDMA extracts from the data.

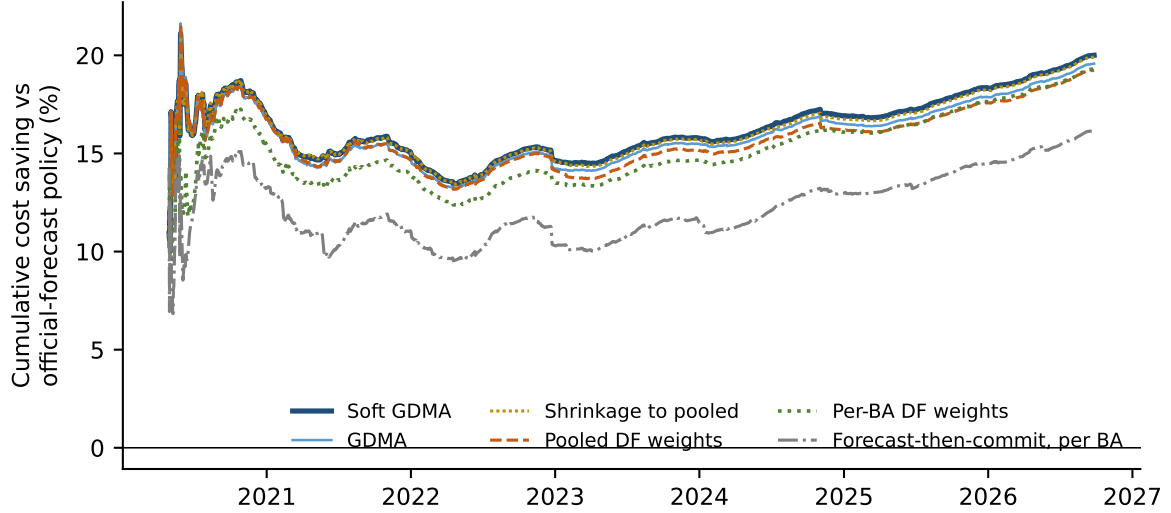


Figure 1: Cumulative system-wide cost saving relative to the official-forecast policy ($W = 28$).

Table 5: Sensitivity to the cost asymmetry ($W = 28$): system cost relative to the official-forecast policy when every BA has the same critical ratio τ , and under the calibrated heterogeneous ratios.

Method	$\tau = 0.5$	$\tau = 0.8$	$\tau = 0.9$	Calibrated
Official forecast	1.0000	1.0000	1.0000	1.0000
Equal weights	1.4030	1.4367	1.4900	1.5322
Best single (per BA)	0.8157	0.7921	0.8081	0.8245
Pooled DF weights	0.7986	0.7775	0.7900	0.8076
Per-BA DF weights	0.8006	0.7798	0.7915	0.8069
Shrinkage to pooled	0.7930	0.7714	0.7837	0.8010
Two-step k -means	0.7949	0.7736	0.7858	0.8041
Forecast-then-commit, per BA	0.8136	0.7944	0.8144	0.8382
Forecast-then-commit, grouped	0.8128	0.7916	0.8066	0.8260
GDMA	0.7965	0.7751	0.7876	0.8042
GDMA, minimum hold-out rule	—	—	—	0.8034
Soft GDMA	—	—	—	0.8000

6.7 Extreme events and robustness

Table 4 examines three extreme episodes. During Winter Storm Uri and the 2023 heat wave, the decision-focused combinations cost 15–22% less than the official-forecast policy, in line with their average performance. Winter Storm Elliott is different: the sudden cold snap of December 2022 was forecast much better by the operators, whose forecasts use weather information that none of our candidates sees directly, and every combination is 18–32% more expensive than the official policy over those six days. Soft GDMA and shrinkage limit the damage better than pooled weights and plain GDMA (1.21 against 1.32), presumably because unit-level shrinkage lets the BAs whose official forecasts are most informative keep more weight on them. The episode is a reminder that combination weights learned in calm weeks cannot anticipate an abrupt regime change; adding weather-based candidates, or letting weights respond to weather warnings, is a natural extension.

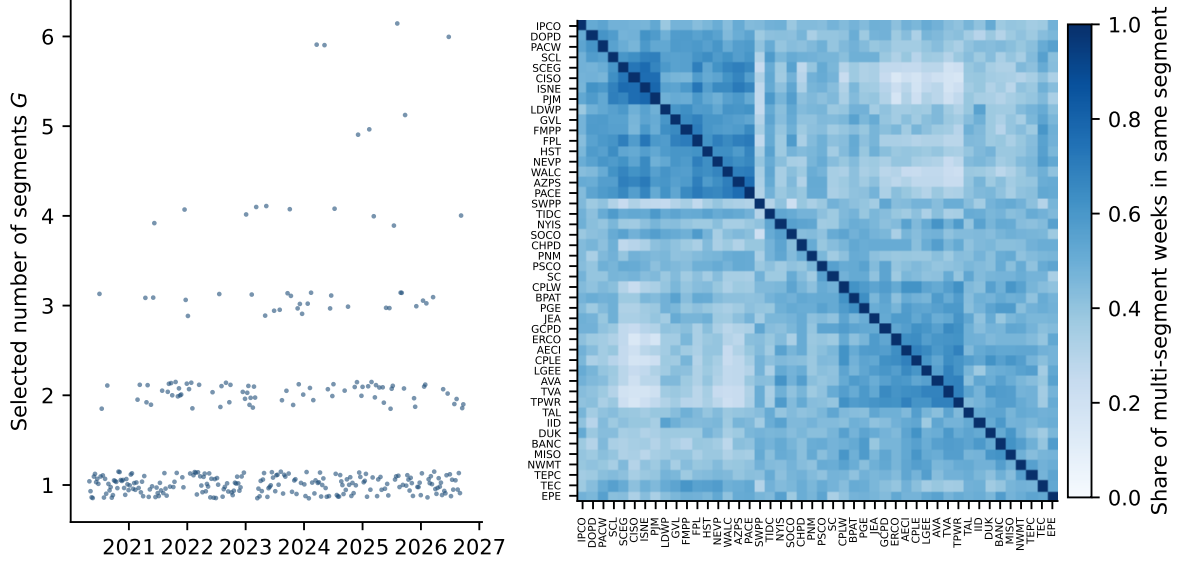


Figure 2: Left: number of segments selected by GDMA in each weekly re-estimation ($W = 28$). Right: among the weeks with $G \geq 2$, share of weeks in which two BAs belong to the same segment, ordered by average-linkage clustering.

7 Managerial implications and conclusion

Implications for system planners. Three lessons emerge for organisations that coordinate many operating units—grid coordinators and regulators, but also national health systems allocating capacity across regions or humanitarian agencies pre-positioning supplies across countries. First, the object to be learned is the *decision rule*, not the forecast: the cost-minimising combination of forecasting systems depends on the cost asymmetry each unit faces, so a combination tuned for accuracy leaves money on the table. Second, planners need not choose between a single national rule and a bespoke rule for every unit. A handful of data-driven segments captures the heterogeneity that matters, is estimated with far less noise than unit-specific rules, and produces a segmentation that can be discussed with operators and regulators—who is grouped with whom, and why. Third, because GDMA only re-weights rules that operators already produce, it can be adopted without replacing existing forecasting systems, and each unit’s commitment remains a convex combination of commitments it already understands.

Limitations and extensions. Our cost calibration is stylised: shortfall costs are tied to a transparent proxy of import flexibility rather than to market prices, and the single-period newsvendor abstracts from ramping, network constraints and multi-day commitments. The timing assumption that the previous day’s demand is fully observed before commitment is generous to persistence-type candidates, although all combination methods share the same information. The theory treats the number of segments as given and relies on a high-level concentration condition; a formal analysis of the hold-out choice of G , and of strong cross-sectional dependence through common weather shocks, is left for future work. Natural extensions include segment-specific weights that vary smoothly with observable states, coupling constraints across units (for example shared reserve pools), and distributionally robust versions of (2) for periods of rapid

structural change.

Conclusion. Grouped decision-focused model averaging combines three ideas that have so far developed separately—optimal model averaging, decision-focused learning and latent group structures—into one estimator with finite-sample and asymptotic guarantees. It gives system planners a principled way to combine the many forecasts they already have into commitment rules that are cheaper, statistically stable and interpretable.

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A Proofs

Throughout, $\|\cdot\|_1$ is the ℓ_1 norm and constants c, C may change from line to line.

Covering the simplex. For $\epsilon \in (0, 1]$ there is a set $\mathcal{N}_\epsilon \subset \Delta_M$ with $|\mathcal{N}_\epsilon| \leq (3/\epsilon)^M$ such that every $\mathbf{w} \in \Delta_M$ has a point $\mathbf{w}' \in \mathcal{N}_\epsilon$ with $\|\mathbf{w} - \mathbf{w}'\|_1 \leq \epsilon$ (volumetric argument for the ℓ_1 ball, e.g. Vershynin, 2018, Ch. 4). Hence $\mathcal{N}_\epsilon^G$ is an ϵ -net of Δ_M^G in the norm $\max_g \|\mathbf{w}_g - \mathbf{w}'_g\|_1$, of cardinality at most $(3/\epsilon)^{GM}$.

Lipschitz property. For all $(\gamma, \mathbf{W})$ and $(\gamma, \mathbf{W}')$ with $\max_g \|\mathbf{w}_g - \mathbf{w}'_g\|_1 \leq \epsilon$, Assumption 1 gives $|\hat{R}(\gamma, \mathbf{W}) - \hat{R}(\gamma, \mathbf{W}')| \leq \bar{a}B\epsilon$ and $|R(\gamma, \mathbf{W}) - R(\gamma, \mathbf{W}')| \leq \bar{a}B\epsilon$.

Proof of Theorem 1

Fix $\epsilon = 1/(NT_e)$. The set $\mathcal{F} = \{1, \dots, G\}^N \times \mathcal{N}_\epsilon^G$ has cardinality at most $G^N(3NT_e)^{GM}$. By Assumption 2 and the union bound,

$$\Pr \left(\max_{(\gamma, \mathbf{W}) \in \mathcal{F}} |\hat{R}(\gamma, \mathbf{W}) - R(\gamma, \mathbf{W})| > x \right) \leq 2G^N(3NT_e)^{GM} \exp(-cNT_e x^2/L^2).$$

Setting the right-hand side equal to δ gives, with probability at least $1 - \delta$, $\max_{\mathcal{F}} |\hat{R} - R| \leq x_\delta := L\{(GM \log(3NT_e) + N \log G + \log(2/\delta))/(cNT_e)\}^{1/2}$. For an arbitrary $(\gamma, \mathbf{W}) \in \mathcal{P}_G$ let $\mathbf{W}'$ be its nearest net point; by the Lipschitz property,

$$\sup_{\mathcal{P}_G} |\hat{R} - R| \leq x_\delta + 2\bar{a}B\epsilon.$$

Let $(\gamma^*, \mathbf{W}^*)$ satisfy $R(\gamma^*, \mathbf{W}^*) \leq R_G^* + \eta$ for arbitrary $\eta > 0$. Since $(\hat{\gamma}, \hat{\mathbf{W}})$ minimises $\hat{R}$,

$$R(\hat{\gamma}, \hat{\mathbf{W}}) \leq \hat{R}(\hat{\gamma}, \hat{\mathbf{W}}) + x_\delta + 2\bar{a}B\epsilon \leq \hat{R}(\gamma^*, \mathbf{W}^*) + x_\delta + 2\bar{a}B\epsilon \leq R_G^* + \eta + 2x_\delta + 4\bar{a}B\epsilon.$$

Letting $\eta \downarrow 0$ and substituting $\epsilon = 1/(NT_e)$ gives (4). $\square$

Proof of Corollary 1

Theorem 1 with $\delta = \delta_{N,T} \rightarrow 0$ slowly (for instance $\log(1/\delta) = \log(NT_e)$) shows $R(\hat{\gamma}, \hat{\mathbf{W}}) - R_G^* = O_p(r_{N,T})$ with $r_{N,T} = \{GM \log(NT_e)/(NT_e) + \log G/T_e + \log(NT_e)/(NT_e)\}^{1/2} \rightarrow 0$. Dividing by $R_G^* \geq \kappa$ yields $R(\hat{\gamma}, \hat{\mathbf{W}})/R_G^* - 1 = O_p(r_{N,T}/\kappa) \rightarrow_p 0$. Because $R(\hat{\gamma}, \hat{\mathbf{W}}) \geq R_N^*$ always and $R_G^* \geq R_N^*$, the second claim follows once we show $R_G^* = R_N^*$ under the stated structure. Let $\mathbf{w}_g^0$ be a common minimiser of R_i for all i with $\gamma_i^0 = g$ (padding with arbitrary vectors if $G_0 < G$); then $R(\gamma^0, \mathbf{W}^0) = N^{-1} \sum_i \min_{\mathbf{w}} R_i(\mathbf{w}) = R_N^*$, so $R_G^* \leq R_N^*$, and the reverse inequality holds because $\mathcal{P}_G$ is a subset of the unit-specific class. $\square$

Proof of Theorem 2

Step 1 (average consistency). Under Assumption 3(a), $R_G^* = R(\gamma^0, \mathbf{W}^0) = R_N^*$ and

$$\frac{\lambda}{N} \sum_{i=1}^N \|\hat{\mathbf{w}}_{\hat{\gamma}_i} - \mathbf{w}_{\gamma_i^0}^0\|_2^2 \leq R(\hat{\gamma}, \hat{\mathbf{W}}) - R_G^* = o_p(1)$$

by Theorem 1, since $\log N/T_e \rightarrow 0$ implies $\log G/T_e \rightarrow 0$ and $GM \log(NT_e)/(NT_e) \rightarrow 0$ for fixed G and M .

Step 2 (consistency of the segment weights). Fix $\eta \in (0, d/4)$. By Step 1 and Markov's inequality, the fraction of units with $\|\hat{\mathbf{w}}_{\hat{\gamma}_i} - \mathbf{w}_{\gamma_i^0}^0\|_2 > \eta$ is $o_p(1)$. Since every true segment g contains at least πN units, with probability approaching one each g contains a unit i with $\|\hat{\mathbf{w}}_{\hat{\gamma}_i} - \mathbf{w}_g^0\|_2 \leq \eta$; call $h(g) = \hat{\gamma}_i$. If $h(g) = h(g')$ for $g \neq g'$, the triangle inequality gives

$\|\mathbf{w}_g^0 - \mathbf{w}_{g'}^0\|_2 \leq 2\eta < d$, contradicting Assumption 3(b). Thus h is injective, hence a bijection of $\{1, \dots, G\}$, and after relabelling $\max_g \|\hat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \leq \eta$ with probability approaching one. Since η is arbitrary, (i) follows.

Step 3 (memberships). On the event $E_\eta = \{\max_g \|\hat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \leq \eta\}$, unit i with $\gamma_i^0 = g$ can be assigned to $h \neq g$ only if $\hat{R}_i(\hat{\mathbf{w}}_h) \leq \hat{R}_i(\hat{\mathbf{w}}_g)$; every global minimiser assigns each unit to a segment minimising its own empirical cost, since otherwise reassigning the unit would lower (2). By Assumption 3 and the Lipschitz property (with $\|\cdot\|_1 \leq \sqrt{M}\|\cdot\|_2$),

$$R_i(\hat{\mathbf{w}}_h) - R_i(\hat{\mathbf{w}}_g) \geq \lambda(d - \eta)^2 - \bar{a}B\sqrt{M}\eta \geq \lambda d^2/4 =: \Delta$$

for η small enough. Hence misclassification of unit i on E_η requires $\sup_{\mathbf{w} \in \Delta_M} |\hat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| \geq \Delta/2$. Covering Δ_M by $\mathcal{N}_\epsilon$ with $\epsilon = \Delta/(8\bar{a}B)$ and applying the unit-level part of Assumption 2,

$$\Pr\left(\sup_{\mathbf{w}} |\hat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| \geq \Delta/2\right) \leq 2\left(\frac{24\bar{a}B}{\Delta}\right)^M \exp\left(-\frac{cT_e\Delta^2}{16L^2}\right) =: C_M e^{-c'T_e}.$$

Because the supremum is over all $\mathbf{w}$, the bound does not require independence between $\hat{\mathbf{W}}$ and unit i 's data. A union bound over units gives $\Pr(\hat{\gamma} \neq \gamma^0) \leq \Pr(E_\eta^c) + NC_M e^{-c'T_e}$, which tends to zero because $\log N/T_e \rightarrow 0$. This is (ii).

Step 4 (oracle equivalence). On the event $\{\hat{\gamma} = \gamma^0\}$, $\hat{\mathbf{W}}$ minimises $\hat{R}(\gamma^0, \cdot)$, so $\hat{\mathbf{W}} = \tilde{\mathbf{W}}$ (if the minimiser is not unique, the two estimators select from the same set). By (ii) this event has probability approaching one. $\square$

B Out-of-sample risk

The theory concerns the population risk R evaluated at the estimated policy. In a rolling application the policy estimated on periods $1, \dots, T$ is applied in periods $T+1, \dots, T+k$. If $\{z_{it}\}$ is β -mixing with coefficients $\beta(\cdot)$, a coupling argument (Yu, 1994) shows that the expected cost of the estimated policy in period $T+s$ differs from $\mathbb{E}R(\hat{\gamma}, \hat{\mathbf{W}})$ by at most $L\beta(s)$. The bounds of Section 4 therefore also bound the out-of-sample cost up to a term that vanishes with the gap between estimation and application.

C Balancing authorities

Table 6: The 46 balancing authorities, ordered by mean hourly demand over 2019–2026. Flex.: interchange flexibility in 2019 (range between the 5th and 95th percentiles of hourly total interchange, divided by mean demand). τ_i : calibrated critical ratio.

Code	Balancing authority	Mean MW	Flex.	τ_i
PJM	PJM Interconnection, LLC	92,501	0.06	0.947
MISO	Midcontinent Independent System Operator, Inc.	74,187	0.07	0.943
ERCO	Electric Reliability Council of Texas, Inc.	49,746	0.03	0.950
SWPP	Southwest Power Pool	32,259	0.08	0.940
SOCO	Southern Company Services, Inc. - Trans	27,072	0.12	0.927
CISO	California Independent System Operator	25,318	0.29	0.900
TVA	Tennessee Valley Authority	18,636	0.09	0.937
NYIS	New York Independent System Operator	17,305	0.18	0.910
FPL	Florida Power & Light Co.	15,870	0.10	0.933
ISNE	ISO New England	13,227	0.16	0.920
DUK	Duke Energy Carolinas	12,210	0.23	0.907
CPLE	Duke Energy Progress East	7,044	0.37	0.880
BPAT	Bonneville Power Administration	6,631	0.84	0.827
PACE	PacifiCorp East	5,803	0.42	0.877
PSCO	Public Service Company of Colorado	5,312	0.11	0.930
NEVP	Nevada Power Company	4,522	0.33	0.883
LGEE	Louisville Gas and Electric Company and Kentucky Utilities Company	4,166	0.17	0.913
AZPS	Arizona Public Service Company	3,893	0.67	0.843
SC	South Carolina Public Service Authority	2,901	0.32	0.890
LDWP	Los Angeles Department of Water and Power	2,861	0.30	0.897
SCEG	Dominion Energy South Carolina, Inc.	2,814	0.15	0.923
AECI	Associated Electric Cooperative, Inc.	2,658	0.52	0.867
TEC	Tampa Electric Company	2,530	0.26	0.903
PGE	Portland General Electric Company	2,523	0.53	0.863
PACW	PacifiCorp West	2,392	0.55	0.860
FMPP	Florida Municipal Power Pool	2,135	0.17	0.917
IPCO	Idaho Power Company	2,097	0.79	0.833
BANC	Balancing Authority of Northern California	1,993	0.59	0.853
PNM	Public Service Company of New Mexico	1,645	0.64	0.847
TEPC	Tucson Electric Power	1,590	0.71	0.837
JEA	JEA	1,532	0.32	0.893
AVA	Avista Corporation	1,460	0.70	0.840
NWMT	NorthWestern Corporation	1,358	1.07	0.810
SCL	Seattle City Light	1,076	0.92	0.817
WALC	Western Area Power Administration - Desert Southwest Region	1,045	0.95	0.813
EPE	El Paso Electric Company	1,029	0.46	0.873
GCPD	Public Utility District No. 2 of Grant County, Washington	696	1.58	0.807
CPLW	Duke Energy Progress West	564	0.61	0.850
TPWR	City of Tacoma, Department of Public Utilities, Light Division	540	0.56	0.857
IID	Imperial Irrigation District	441	0.91	0.820
TIDC	Turlock Irrigation District	328	0.81	0.830
TAL	City of Tallahassee	322	0.32	0.887
DOPD	PUD No. 1 of Douglas County	245	2.20	0.803
GVL	Gainesville Regional Utilities	240	0.50	0.870
CHPD	Public Utility District No. 1 of Chelan County	218	4.70	0.800
HST	City of Homestead	71	0.90	0.823